STACKS, DISCRIMINANTS: LEMMA-DEDEKIND-DIFFERENT-IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1656-1668 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1670-1692, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 25 of queue_discriminant.json (pb=1023, stmt 1656-1668, proof 1670-1692).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00H7: sources/stacks-discriminant/pinned-original-source.tex, source lines 1692–1692. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BIH: sources/stacks-discriminant/pinned-original-source.tex, source lines 1692–1692. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Assume the Dedekind different of $A \to B$ is defined. Consider the statements*

(1) *$A \to B$ is flat,*
(2) *A is a normal ring,*
(3) *$\mathit{Trace}_{L/K}(B) \subset A$,*
(4) *$1 \in \mathcal{L}_{B/A}$, and*
(5) *the Dedekind different $\mathfrak{D}_{B/A}$ is an ideal of B.*

Then we have (1) $\Rightarrow$ (3), (2) $\Rightarrow$ (3), (3) $\Leftrightarrow$ (4), and (4) $\Rightarrow$ (5).

Proof. The equivalence of (3) and (4) and the implication (4) $\Rightarrow$ (5) are immediate.

If $A \to B$ is flat, then we see that $\mathrm{Trace}_{B/A} : B \to A$ is defined and that $\mathrm{Trace}_{L/K}$ is the base change. Hence (3) holds.

If A is normal, then A is a finite product of normal domains, hence we reduce to the case of a normal domain. Then K is the fraction field of A and $L = \prod L_i$

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is a finite product of finite separable field extensions of K. Then $\mathrm{Trace}_{L/K}(b) = \sum \mathrm{Trace}_{L_i/K}(b_i)$ where $b_i \in L_i$ is the image of b. Since b is integral over A as B is finite over A, these traces are in A. This is true because the minimal polynomial of b_i over K has coefficients in A (Algebra, Lemma 00H7) and because $\mathrm{Trace}_{L_i/K}(b_i)$ is an integer multiple of one of these coefficients (Fields, Lemma 0BIH). $\square$